# Part A: Conceptual Foundation (Theory) - Answers

## 1. What are Ensemble Learning Methods and why are they effective?

Ensemble Learning methods combine multiple machine learning models to make a final prediction. They improve accuracy, reduce errors, and provide more reliable and robust predictions than a single model.

## 2. Explain the difference between Bagging, Boosting, and Voting.

Bagging trains multiple models independently on random subsets of data and combines their outputs. Boosting trains models sequentially, where each model corrects previous mistakes. Voting combines predictions from different models using majority voting or probability averaging.

## 3. What is Bias–Variance Trade-off, and how do ensemble methods address it?

The Bias–Variance Trade-off is the balance between underfitting (high bias) and overfitting (high variance). Bagging reduces variance, while Boosting reduces bias. Ensemble methods help achieve better overall model performance.

## 4. Explain Voting Classifier and Stacking Ensemble.

A Voting Classifier combines predictions from multiple models using hard or soft voting. Stacking uses several base models and a meta-model that learns from their predictions to produce a final prediction.

## 5. Compare AdaBoost, Gradient Boosting, LightGBM, and XGBoost conceptually.

AdaBoost focuses on misclassified samples by increasing their weights. Gradient Boosting minimizes prediction errors using gradients. XGBoost is an optimized version of Gradient Boosting with regularization and speed improvements. LightGBM is a fast and memory-efficient boosting framework designed for large datasets.